20 50 people were asked if they speak French or German or Spanish.

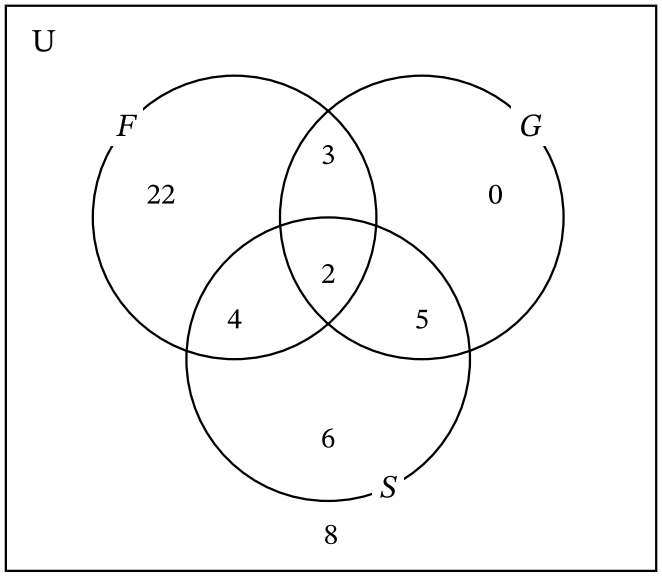
Of these people,

31 speak French
2 speak French, German and Spanish
4 speak French and Spanish but not German
7 speak German and Spanish
8 do not speak any of the languages
all 10 people who speak German speak at least one other language

Two of the 50 people are chosen at random.

Work out the probability that they both only speak Spanish.

F = French; G = German; S = Spanish. U is all 50 people.



G and S only: 7 – 2 = 5.

G only = 0.

F and G only: 10 – 5 – 2 = 3.

F only: 31 – 3 – 4 – 2 = 22.

1 mark
Find a first unknown region

1 mark
Find at least three unknown regions

Spanish only = 50 – (8 + 22 + 3 + 0 + 4 + 2 + 5) = 6.

1 mark
Account for all the other disjoint regions

Choose without replacement: both must be from these six people.

$$\text{Probability} = \frac{6}{50} \times \frac{5}{49} = \frac{30}{2450} = \frac{3}{245}$$

1 mark
Use 6 out of 50, then 5 out of the remaining 49

1 mark
Correct probability

$\frac{3}{245}$

(Total for Question 20 is 5 marks)